Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an AI- and LLM-Based Multi-Sensor Fusion Intelligent Transport and Conversational Location-Tracking System for Industrial High-Pressure Gas Cylinders**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Information and Communication Technology*	
	Medium Category	Small Category	Medium Category	Small Category
	Smart Manufacturing	*Industrial Material Handling and Tracking Systems*	*Artificial Intelligence / Data Intelligence*	*Multi-Modal Sensor Fusion AI and LLM-based Decision Systems*
Concept of Proposed Technology	**Core Concept** This project aims to develop an intelligent gas cylinder management system that overcomes existing limitations in tracking reliability and safety management throughout the transportation, storage, and usage lifecycle of industrial gas cylinders. The project seeks to establish an integrated management platform that enables high-reliability tracking and proactive risk prevention across industrial gas supply chains, semiconductor and chemical manufacturing processes, hydrogen infrastructure, and smart factory environments. **Key Functions and Potential Applications** The proposed system integrates heterogeneous sensors—including RFID, BLE, vision, UWB, and			

	GPS—and combines AI-based sensor fusion with LLM-driven decision-making technologies. The system provides real-time cylinder identification and location tracking, digital twin–based logistics and safety monitoring, and LLM-enabled natural language query responses and interpretation of safety regulation–based alerts.
Rationale for Joint R&D Project	**Critical Technical Barriers** The industrial gas cylinder management environment presents complex challenges that cannot be addressed by a single technology due to metal interference, hazardous material regulations, and intricate operational workflows. This project requires the integration of AI-based perception technologies, sensor fusion algorithms, LLM-based knowledge reasoning, real-world operational data, and safety protocols, making it difficult for a single organization to execute independently. **Why Global Collaboration?** In this joint R&D framework, the research institute will lead the development of core AI and LLM technologies, while the industrial partner will provide real operational environments, datasets, and system demonstration infrastructure. This collaboration enables simultaneous achievement of technological innovation and practical field applicability. **Expected Impact** Through joint research, the project will enhance the feasibility of technology commercialization and strengthen global competitiveness in smart manufacturing and hazardous-material logistics management. **Figure. Industrial Site and Gas Cylinder Handling Status of Modern Korea Co., Ltd.**

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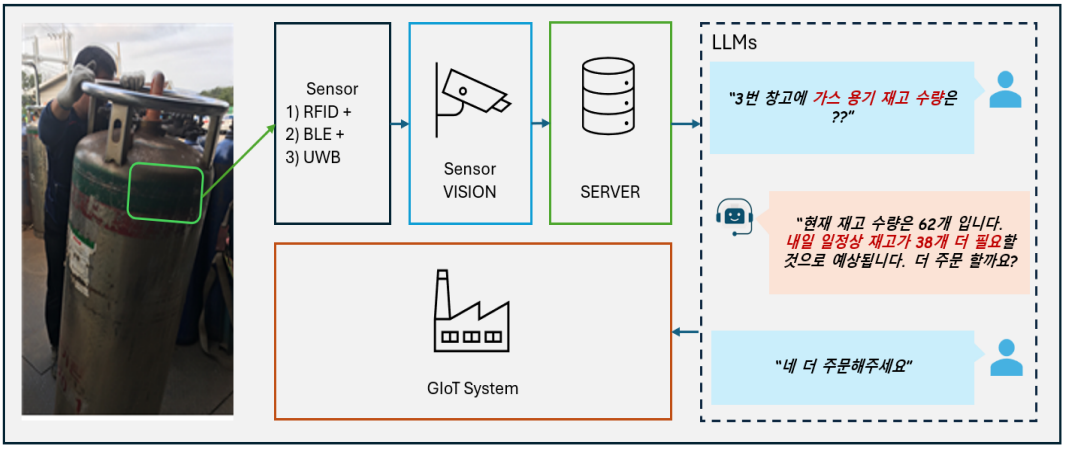

Figure. ***System Architecture Diagram***

Key Words	1. Multi-Sensor Fusion 2. Industrial Gas Cylinder Tracking 3. AI-based Perception 4. LLM-based Decision Support 5. Smart Safety Management

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)

R&D Objectives

R&D Objectives

This project aims to develop an automated system for real-time tracking, logistics optimization, and safety management of industrial gas cylinders by integrating multi-sensor fusion technologies—such as RFID, BLE, vision, and UWB—with AI-based perception algorithms and LLM-driven intelligent decision-making.

Ultimately, the project seeks to implement a pilot-scale integrated prototype that can be operated in real industrial environments, and to validate high-reliability tracking and proactive safety management functions, thereby establishing a foundation for commercialization.

R&D Target Specifications

Item	Target Performance
Cylinder recognition accuracy	≥ 99%
Location tracking error	≤ 30 cm (UWB-based), ≤ 1 m (BLE-based)
Real-time tracking latency	≤ 1 second
System availability	≥ 99%
LLM query response time	≤ 3 seconds

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R&D Plan	This project adopts a phased development approach—sensor hardware deployment → AI-based perception development → LLM-driven decision-making module development → integrated prototype implementation → field demonstration—to advance the technology readiness level (TRL) from TRL3 to TRL6. **Year 1 (Months 1–12): Core Technology Development and System Design** ● Analyze industrial gas cylinder logistics processes and safety requirements ● Design deployment of RFID, BLE, vision, and UWB sensors and establish communication architecture ● Develop multi-sensor data acquisition and synchronization framework ● Develop AI-based cylinder recognition and tracking algorithms ● Design overall system architecture and digital twin models Annual Milestones: ● Completion of system requirement specification ● Laboratory validation of multi-sensor fusion perception algorithms (Achievement of TRL4) **Year 2 (Months 13–24): AI–LLM Integration and Prototype Development** ● Enhance multi-sensor fusion algorithms for scalable operation ● Train and optimize vision-based cylinder recognition and OCR models ● Construct knowledge bases based on industrial safety regulations and operational procedures ● Develop LLM-based modules for natural language queries, alert interpretation, and decision-making ● Implement digital twin–based logistics and safety monitoring systems ● Develop integrated prototype and conduct indoor testing Annual Milestones: ● Completion of AI–LLM integrated intelligent prototype ● Performance validation of integrated system in indoor environments (Achievement of TRL5) **Year 3 (Months 25–36): Field Pilot Demonstration and Commercialization Preparation** ● Deploy sensor networks in real industrial gas cylinder handling sites ● Install and operate integrated prototype for real-time field demonstration ● Evaluate tracking accuracy, system stability, and safety management performance ● Optimize system based on user feedback and operational data ● Establish certification, standardization response, and commercialization roadmap Annual Milestones: ● Completion of pilot-scale field demonstration (Achievement of TRL6) ● Verification of commercialization feasibility and foundation for advancement toward TRL7

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will conduct R&D on AI- and LLM-based intelligent perception and decision-making technologies that form the backbone of the proposed system. 1) The main roles and responsibilities ● Development of multi-sensor fusion–based cylinder recognition and tracking algorithms ● Research and advancement of AI-based vision recognition and abnormal situation detection models ● Development of LLM-based modules for natural language queries, alert interpretation, and decision-making ● Construction of knowledge bases based on industrial safety regulations and operational procedures ● Design of digital twin–based logistics and safety monitoring platforms ● Overall system architecture design and technical integration management ● Dataset construction and analysis for algorithm training and validation ● Establishment of performance evaluation methodologies and quantitative performance verification ● Provision of research personnel and AI–LLM development infrastructure 2) Provided Resources ● AI and LLM research personnel and algorithm development environment ● High-performance GPU servers and data processing infrastructure ● Software toolchains for AI model training and validation ● Technical documentation of R&D outcomes and patent applications
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for field-oriented technology development, system implementation, and pilot demonstration in real industrial environments. It will carry out system deployment, validation, and commercialization in actual industrial gas cylinder handling sites. 1) The main roles and responsibilities are as follows ● Design and installation of sensor hardware, including RFID, BLE, vision, and UWB systems ● Construction and operation of field data acquisition systems ● Provision of industrial gas cylinder logistics, storage, and operational processes ● Provision and operational support of pilot demonstration environments at industrial sites ● Fabrication of integrated prototypes and on-site system installation ● Execution of system demonstration tests and reliability evaluations ● Provision of industrial safety regulations and operational manuals ● Design of commercial products and promotion of commercialization activities ● Provision of on-site operational personnel and equipment 2) Provided Resources

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	● Industrial gas cylinder handling facilities and operational data ● Sensor devices and communication infrastructure ● Personnel for pilot demonstration operation and on-site support systems ● Product manufacturing and system integration development staff

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** Through this project, core technologies will be secured, including multi-sensor fusion–based high-reliability perception integrating RFID, BLE, vision, and UWB sensors; AI-based real-time tracking and abnormal situation detection; and LLM-driven intelligent decision-making and natural language interfaces. These technologies will overcome the recognition gaps, location inaccuracies, and field application limitations of conventional single-RFID-based systems, thereby enabling the digital transformation of the entire logistics and safety management lifecycle of industrial gas cylinders. In particular, by integrating AI perception, digital twin technology, and LLM-based decision-making, the project will establish an intelligent management framework that serves as a platform technology expandable to various industrial domains, including smart factories, hazardous-material logistics, and energy and hydrogen infrastructure. **Commercialization Potential** The developed system can be evolved into a commercially deployable solution applicable to domestic and global industrial sites such as industrial gas suppliers, semiconductor and chemical processes, hydrogen storage and transportation facilities, and gas management systems in hospitals and research institutes. By validating technical reliability and economic feasibility through pilot demonstrations, commercialization can be pursued in the following forms: ● Productization of intelligent gas cylinder management solutions ● System and maintenance service businesses ● Safety management SaaS platform services This will enable early entry into the domestic market and establish a foundation for expansion into global hazardous-material logistics and industrial gas markets. **Industrial and Economic Impact** The adoption of this technology will fundamentally reduce gas cylinder loss, misplacement, overloading, and safety accident risks, while improving logistics operational efficiency and reducing operating costs. Furthermore, automated compliance with industrial safety regulations and proactive accident prevention will contribute to the prevention of accidents and the advancement of industrial safety management. Moreover, AI–LLM–converged smart logistics and safety management technologies align with national policies on smart manufacturing innovation, industrial digital transformation, and carbon neutrality and hydrogen economy initiatives. The project is expected to enhance the development of high value-added AI-based products by domestic SMEs and strengthen global competitiveness.

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